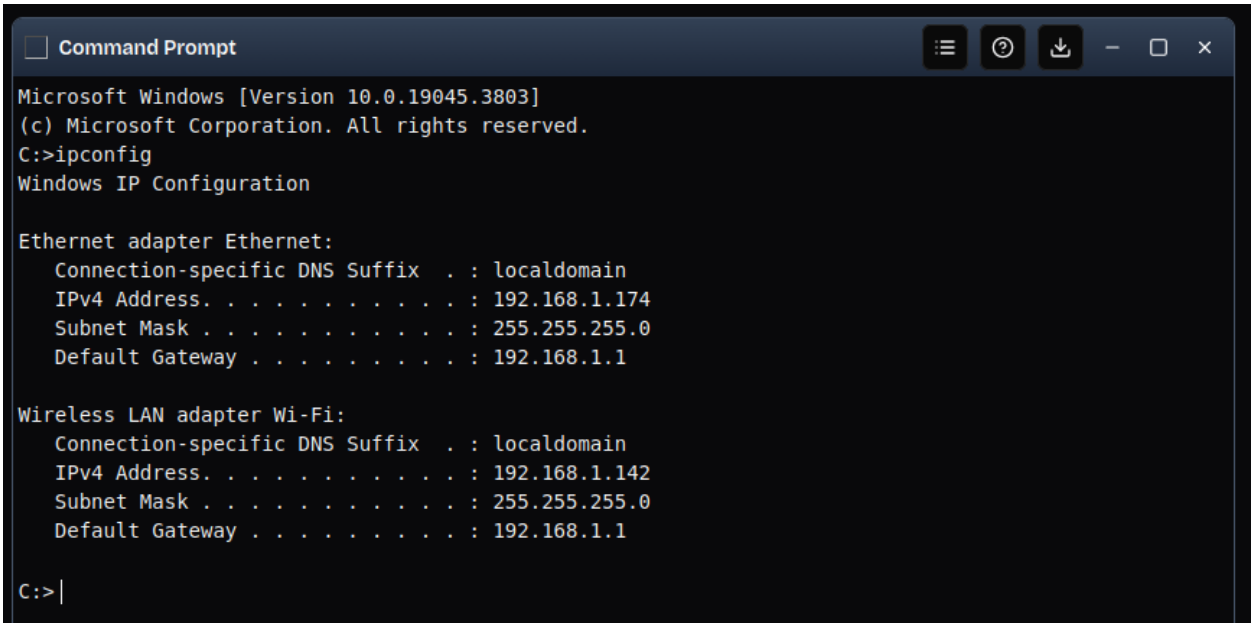
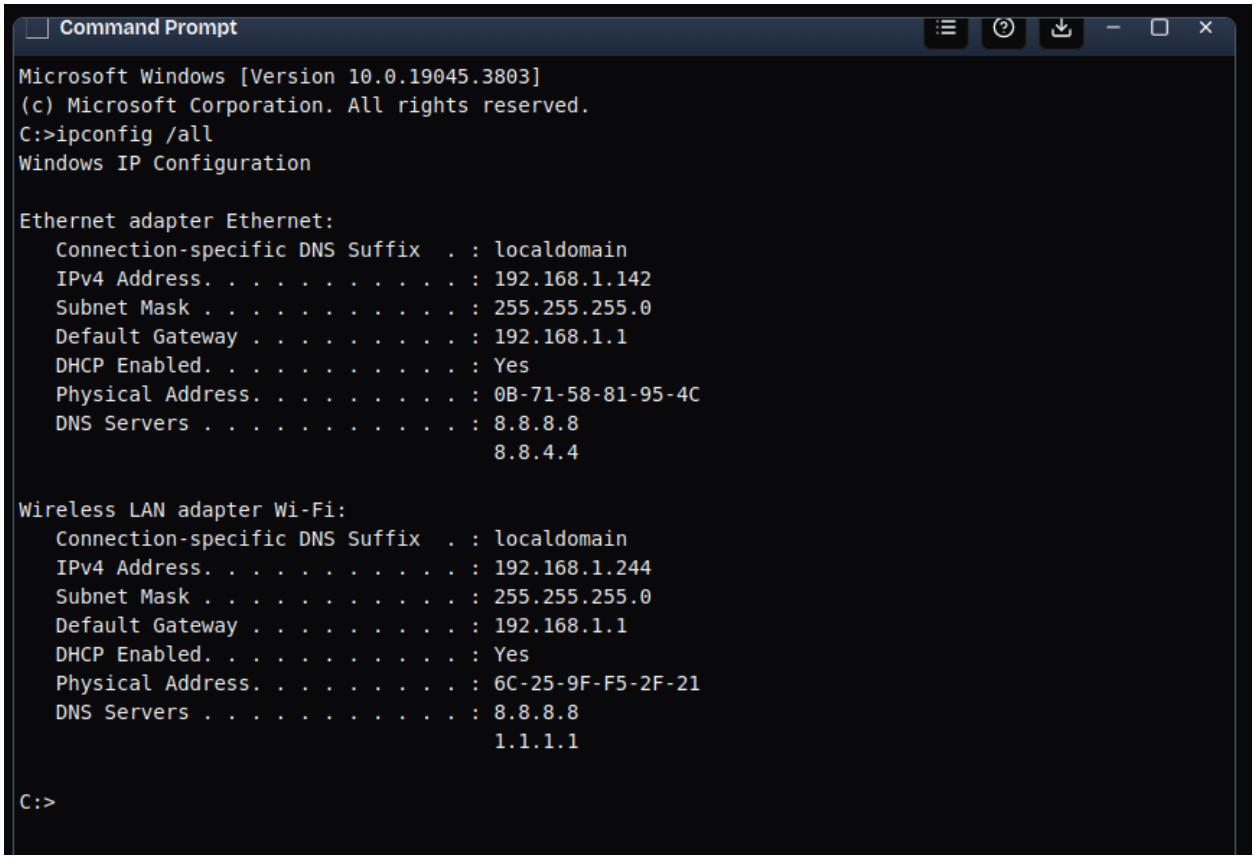
Part 1

1.

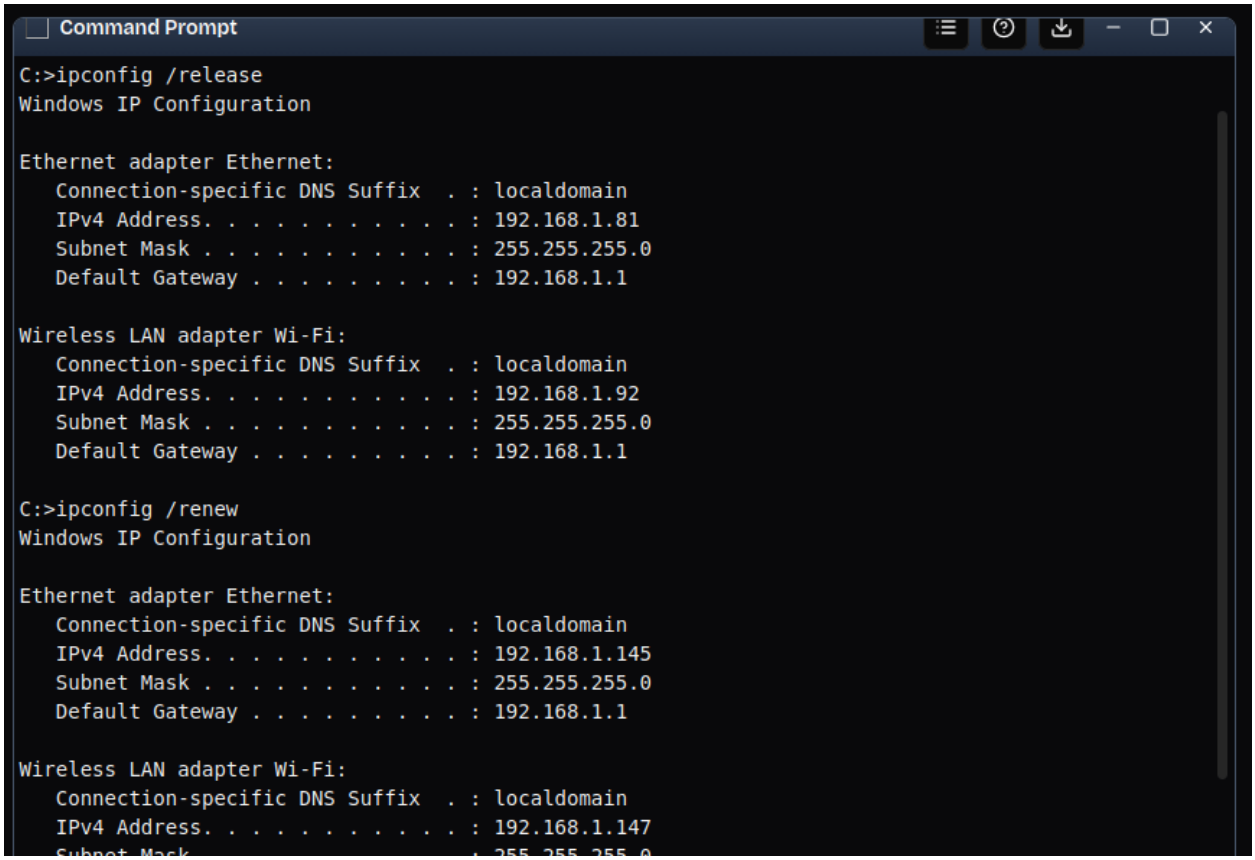

Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : localdomain
IPv4 Address. . . . . : 192.168.1.174
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : localdomain
IPv4 Address. . . . . : 192.168.1.142
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:>

2.


Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>ipconfig /all
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : localdomain
IPv4 Address. . . . . : 192.168.1.142
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Enabled. . . . . : Yes
Physical Address. . . . . : 0B-71-58-81-95-4C
DNS Servers . . . . . : 8.8.8.8
8.8.4.4
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : localdomain
IPv4 Address. . . . . : 192.168.1.244
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Enabled. . . . . : Yes
Physical Address. . . . . : 6C-25-9F-F5-2F-21
DNS Servers . . . . . : 8.8.8.8
1.1.1.1
C:>

3.
IPv4 Address 192.168.1.142
Subnet Mask 255.255.255.0
Defult Gateway 192.168.1.1
DNS server 8.8.8.8
MAC Address 6c-25-9f-f5-2f-21

4.

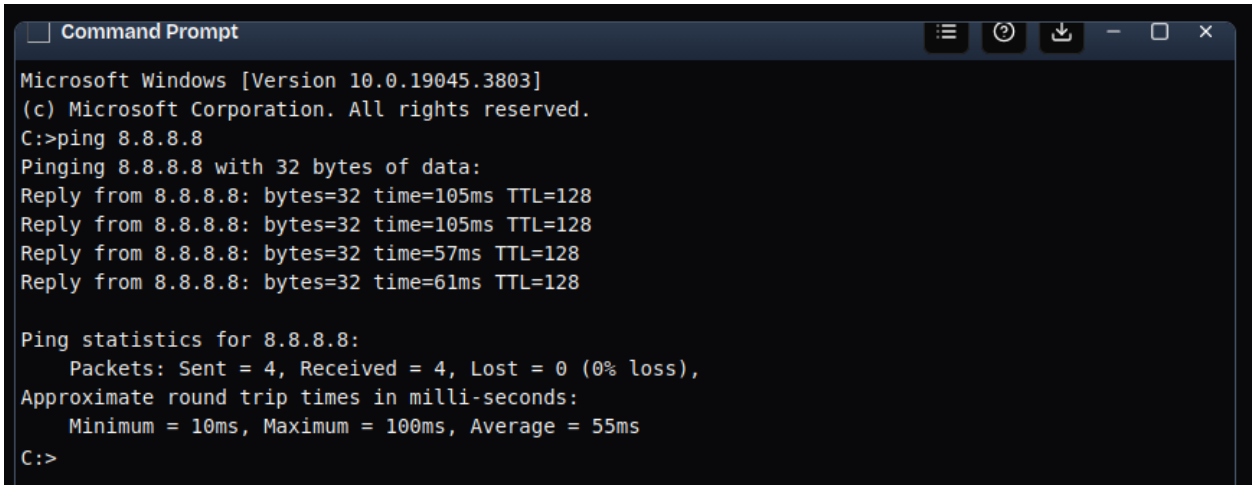


6.
a Static IP address never changes assigned manually
a Dynamic IP address changes automatically over time assigned by router

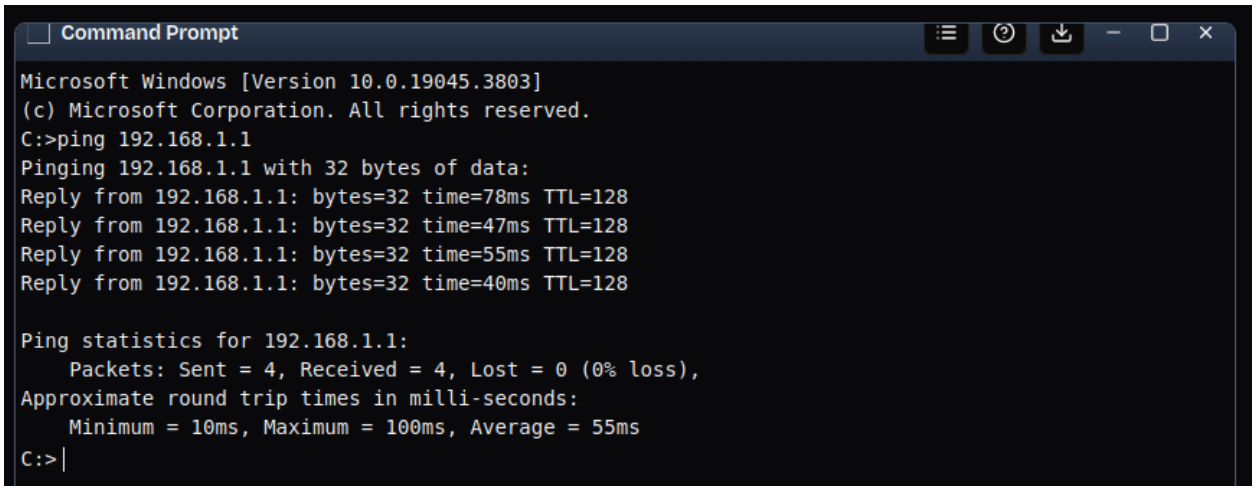
7.
it causes an IP address conflict, which disrupts the network connection for both devices

Part 2

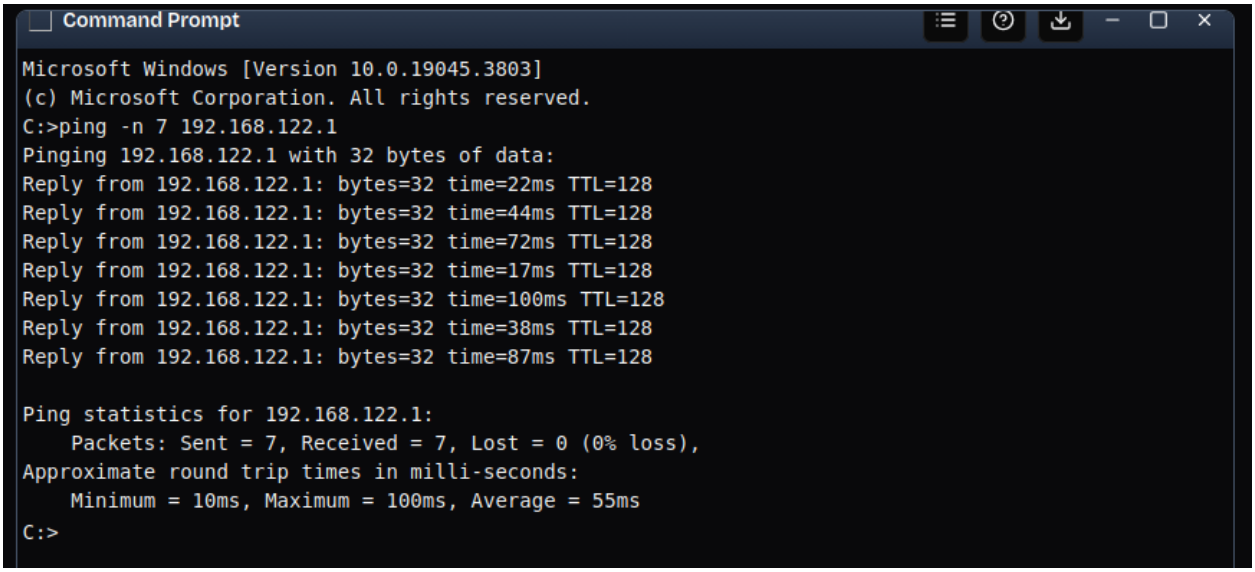
1.


Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=105ms TTL=128
Reply from 8.8.8.8: bytes=32 time=105ms TTL=128
Reply from 8.8.8.8: bytes=32 time=57ms TTL=128
Reply from 8.8.8.8: bytes=32 time=61ms TTL=128
Ping statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 100ms, Average = 55ms
C:>

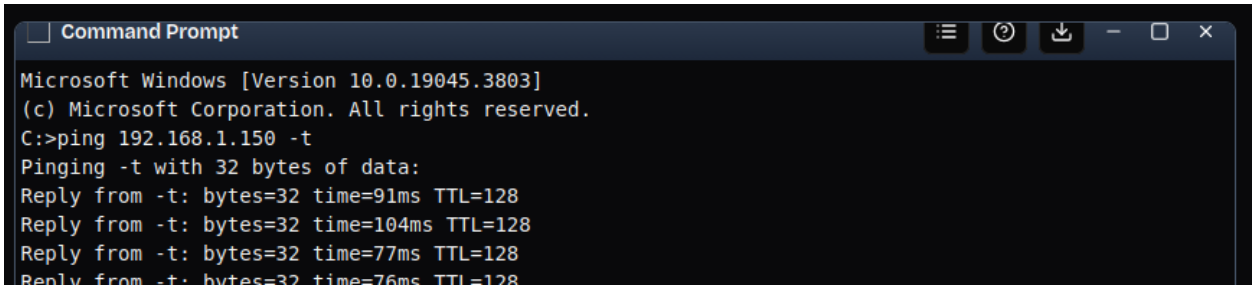
2.


Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=78ms TTL=128
Reply from 192.168.1.1: bytes=32 time=47ms TTL=128
Reply from 192.168.1.1: bytes=32 time=55ms TTL=128
Reply from 192.168.1.1: bytes=32 time=40ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 100ms, Average = 55ms
C:>

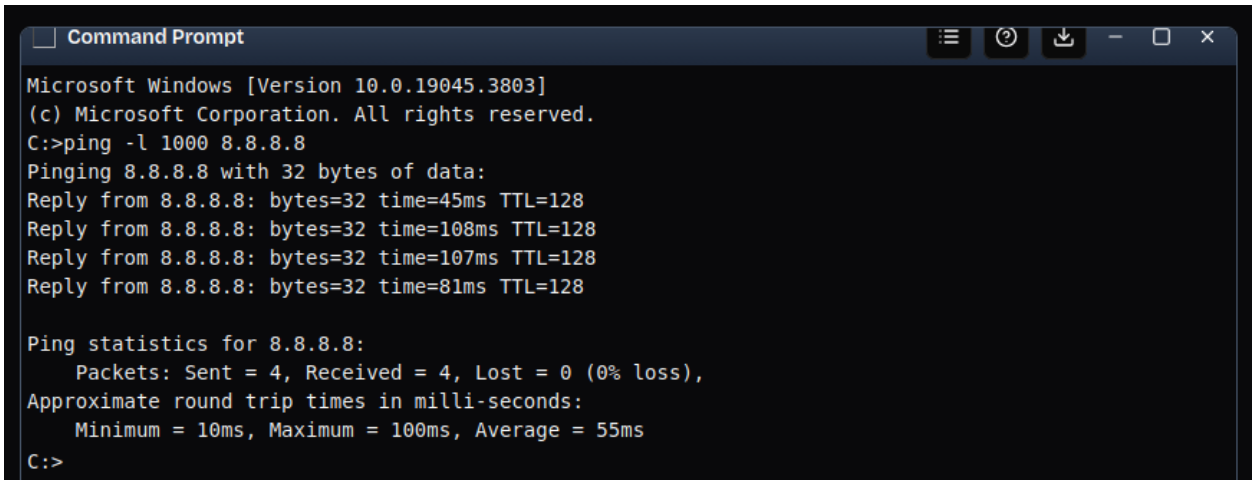
3.


Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:>ping -n 7 192.168.122.1
Pinging 192.168.122.1 with 32 bytes of data:
Reply from 192.168.122.1: bytes=32 time=22ms TTL=128
Reply from 192.168.122.1: bytes=32 time=44ms TTL=128
Reply from 192.168.122.1: bytes=32 time=72ms TTL=128
Reply from 192.168.122.1: bytes=32 time=17ms TTL=128
Reply from 192.168.122.1: bytes=32 time=100ms TTL=128
Reply from 192.168.122.1: bytes=32 time=38ms TTL=128
Reply from 192.168.122.1: bytes=32 time=87ms TTL=128
Ping statistics for 192.168.122.1:
Packets: Sent = 7, Received = 7, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 100ms, Average = 55ms
C:>

4.



5.



6.

"Reply from" means a successful connection

"Request timed out" means a failed connection

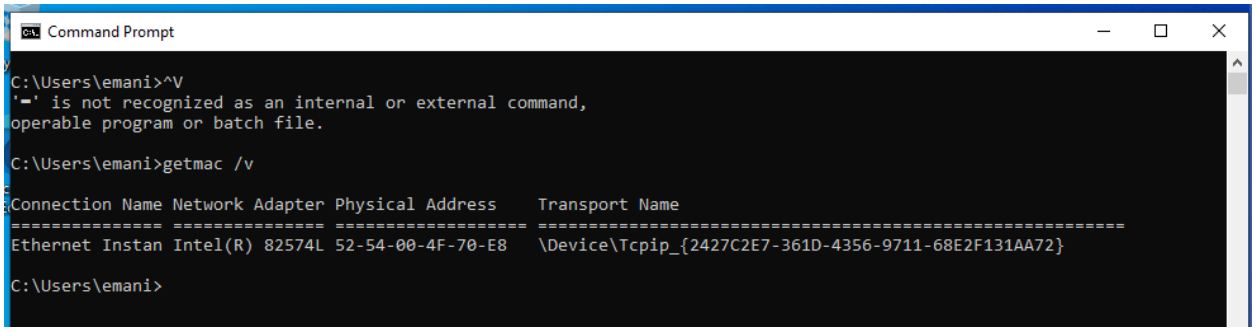
7.

No, a successful Ping does not guarantee that all services on the destination device are working

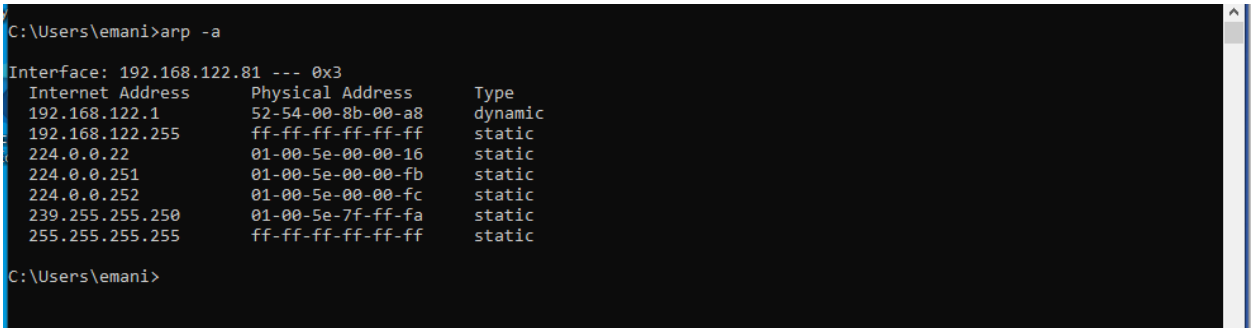
Part 3

1.

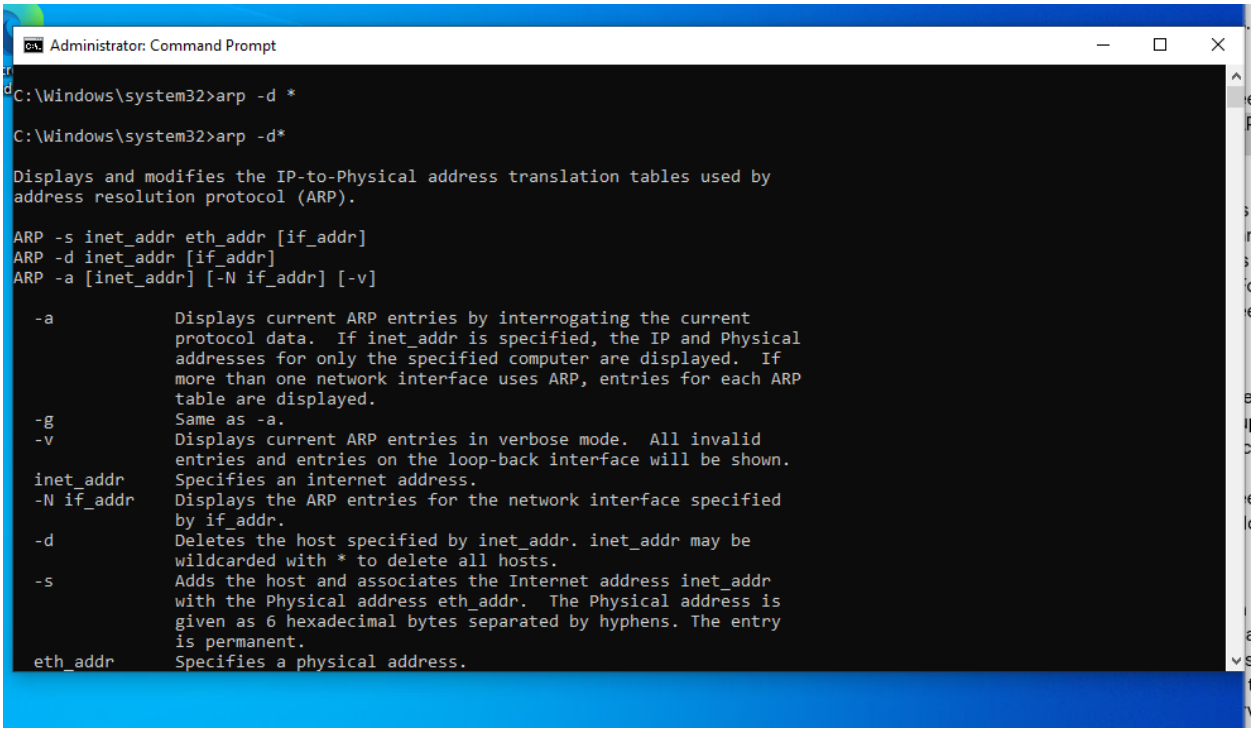
Physical address by ipconfig or getmac /v



2.



3.



4.

IP Address is a temporary, logical address used to route data across different networks (like the internet)

MAC Address is a permanent, physical address used to identify your exact piece of hardware within your local network

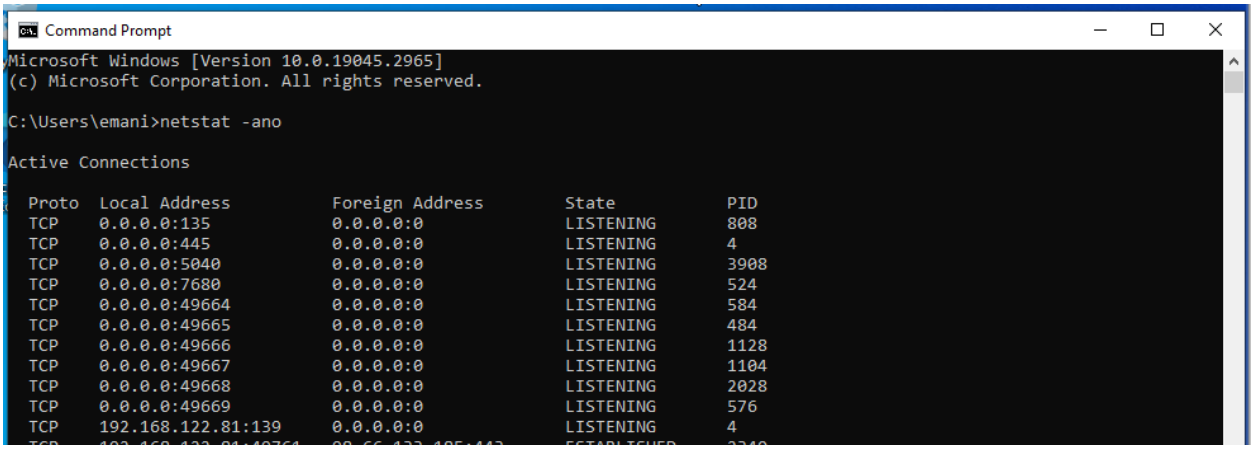
5.

The purpose of the **Address Resolution Protocol (ARP)** is to bridge the gap between network software and network hardware by mapping a known IPv4 address to a physical MAC address on a local network.

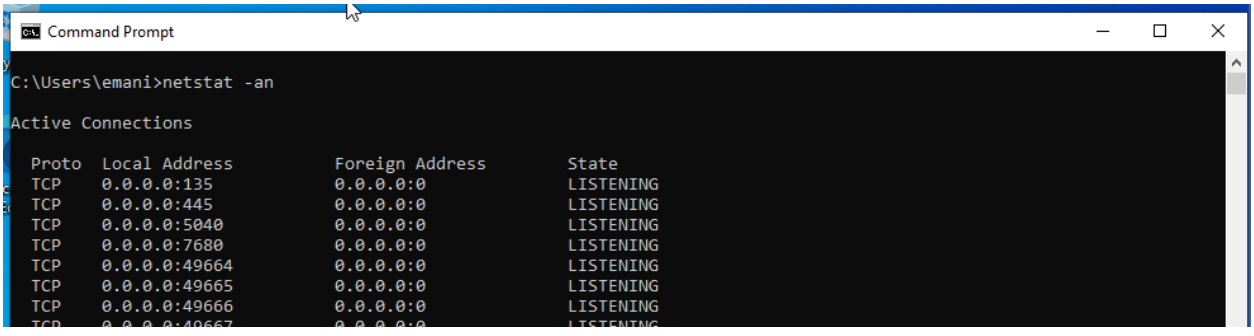
—----------

Part 4

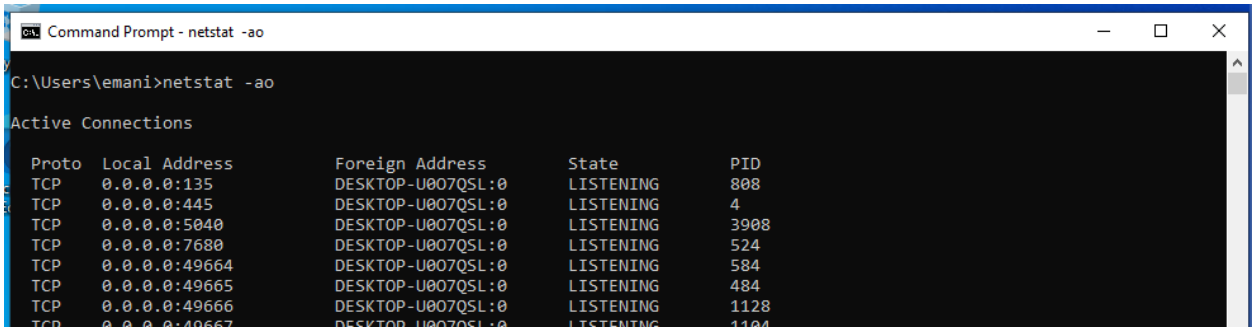
1.



2.



3.



4.

The **Local Address** is your own computer's IP address and the specific port number being used by an application.

The **Foreign Address** (sometimes called Peer Address) is the IP address and port number of the remote computer or server you are communicating with.

The **State** indicates the current status of the network connection:

LISTENING:

Your computer has a port wide open, waiting for an outside connection to come in

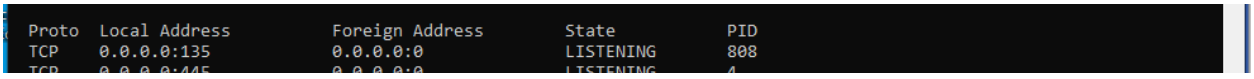
ESTABLISHED:

An active, open data connection is currently happening between your computer and the foreign computer. Data is actively flowing or ready to flow.

TIME_WAIT / CLOSE_WAIT:

The connection is in the process of shutting down or clean-up after data transfer has finished.

The **PID** is a unique number assigned by your operating system to every running program or background service.



5.

LISTENING:

Your computer has a port wide open, waiting for an outside connection to come in

ESTABLISHED:

An active, open data connection is currently happening between your computer and the foreign computer. Data is actively flowing or ready to flow.

Part 1 - 5’

1. Press `Win + R`, type **`ncpa.cpl`**, and press **Enter** to open Network Connections.
2. **Right-click** your active network adapter (Wi-Fi or Ethernet) and select **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** from the list and click the **Properties** button.
4. Click the radio button for **"Use the following IP address"**.
5. Fill in your network details:
    - **IP address:** Choose an IP outside your router's automatic DHCP pool to avoid conflicts (e.g., `192.168.1.150`).
    - **Subnet mask:** Usually populates automatically as `255.255.255.0`.
    - **Default gateway:** Your router's IP address (e.g., `192.168.1.1`).
6. Under DNS, select **"Use the following DNS server addresses"** and enter a public DNS provider like Google (`8.8.8.8`) or Cloudflare (`1.1.1.1`).
7. Click **OK** and then **Close** to apply.

